NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.4a Forecast uses (high-wind background fields, wind power, CFD)

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-13 11:30 — 2026-10-13 13:42

Stage conclusion

Project report

Uses of AI weather foundation models for nowcasting / short-range forecasting

Objective

Evaluate whether data-driven weather models (GraphCast/GenCast, Pangu, Aurora, FuXi, etc.) and AI nowcasting models can provide background fields for tornadoes/thunderstorms/typhoons, serve wind power forecasting, offer utility for CFD, and assess the value of reforecasts for our research. Deliver an assessment table, a literature review audit ID, a minimal experiment, and a one-sentence conclusion. Task brief: (approved by Liang Hu on 10-13).

Current Status

10-13 14:00 **Shelved** (feasibility questions fully answered). Three-event reforecast (Wister, Akron, ALTU): maximum lead time simultaneously satisfying all three pre-specified criteria is 12/12/none for Pangu and 24/none/12 h for FuXi, showing instability; the bottleneck is CAPE (on ALTU, Pangu fails MLCAPE across all five lead times, with errors 1.2–11× that of persistence), whereas wind shear and DCAPE outperform persistence across most combinations. Literature review audit verdict issued. Total compute across both experiment phases is roughly estimated at <11.3 CU (≤20 cap). Figures with source footnotes have been generated: , (previous figures retained).

Reason for shelving: Because we focus on historical event hindcasts, using ERA5/HRRR analysis fields directly for the environment is superior; when using AI global models directly as downburst background fields, the thermodynamic environment only holds up to 12–24 h and is unstable, while 0.25° cannot resolve the storm itself; the bottleneck in the CM1 track lies in observational candidates rather than environmental data. **Missing conditions / Conditions for restart**: (1) Our group pivots toward forecasting tasks (assessing severe convective wind environments several days ahead, or wind power forecasting); (2) The WRF track is willing to test "Pangu-driven WRF vs. GFS-driven WRF" (precedents: Xu 2024 ERL, Liu 2024 JGR:MLC), which is the most promising direction but requires formal project scale; (3) When multi-event statistics are needed, start with existing AWS AIWP reforecasts (from 2022-01 onward, GFS/IFS initial conditions at 00/12 UTC, subject to verifying that pressure-level variables are included); (4) If CAPE is the target, switch to 37-level outputs or a post-processing model (such as the Transformer post-processor in Hua et al., arXiv 2505.11750). Use for restart, ~2–3 CU per event.

Established Decisions

- 2026-10-13 Literature review audit verdict: GraphCast weights open for commercial use starting 2026-08; NowcastNet–HRRR comparison updated to cite Das 2024; reforecast interpretation restricted to qualified scope →
- 2026-10-13 Minimal experiment criteria and conventions (13-level ERA5 as primary reference, diurnally aligned persistence baseline, all three criteria satisfied simultaneously) →
- 2026-10-13 Kickoff plan and minimal experiment direction (multi-initialization reforecasts rather than single-time comparisons) →

Assessment Table (finalized after review verdict, 10-13; recommendation level reflects internal priority judgment)

Question	Our Specific Use Case	Existing Data / Precedents	Minimal Experiment	Cost	Recommendation
1 Severe convective environmental background fields (directly using AI global models)	Assess favorable environments days ahead and decide whether to schedule WRF/CM1 runs; not used for CM1 multi-event screening (CM1 Lead)	Our group's reforecasts; Feldmann 2024 arXiv 2406.09474, Hua 2025 GRL 10.1029/2024GL109611 (abstract)	Completed on three events: under our n=3, domain-maximum, diurnally aligned persistence criteria, maximum usable lead time is Pangu 12/12/none, FuXi 24/none/12 h; wind shear and DCAPE beat persistence in most combinations, while MLCAPE/MUCAPE is the bottleneck	<11.3 CU	Low–Medium
1′ AI large-scale fields driving WRF	Extend usable lead time of WRF driving fields	Xu 2024 ERL 10.1088/1748-9326/ad41f0 (Pangu→WRF heavy rain TS maintained for 8.5 days vs. ~3 days for GFS), Liu 2024 JGR:MLC 10.1029/2024JH000207 (typhoon intensity error reduced by 32% vs. global model and 23% vs. ERA5-driven WRF); CM1 can only ingest a single sounding	Untested, requires coordination with WRF track	—	Medium

Question	Our Specific Use Case	Existing Data / Precedents	Minimal Experiment	Cost	Recommendation
1b Convection-allowing AI and nowcasting	0–12 h thunderstorms and gusts	StormCast arXiv 2408.10958, WoFSCast GRL 10.1029/2024GL112383 (numerical model emulators); FuXi-Nowcast arXiv 2512.08974 (includes gusts); NowcastNet third-party evaluation Das 2024 10.1038/s41612-024-00834-8 (CSI 0.30 vs. HRRR 0.04, >16 mm/h)	Not for now: no CAM training archive or radar training data	—	Medium (Monitor)
1c Typhoons	Track / intensity background fields	DeMaria 2025 AIES 10.1175/AIES-D-24-0085.1 (track comparable to best operational models, intensity systematically biased weak); near-surface wind profiles require coupling with WRF or -style reconstruction (unverified when coupled to AI forecasts)	As needed	—	Medium
2 Wind power	Hub-height wind / power	GenCast Nature 10.1038/s41586-024-08252-9 (97.2%/1320; wind power evaluated via idealized power curve aggregation); Hardy 2025 10.1002/met.70038 (0.18 vs. 0.63 is contemporaneous ERA5 reconstruction; AIFS forecast outperforms IFS after ~Day 3)	Do not pursue (outside our group's main track)	—	Medium–High for wind power itself; Low for our group's main track

Question	Our Specific Use Case	Existing Data / Precedents	Minimal Experiment	Cost	Recommendation
3 CFD	Downscaling chain boundary conditions, inflow	FuXi-CFD Nat Commun 10.1038/s41467-026-70562-5 (30 m terrain wind including TKE, validated against met masts; trained on neutral steady RANS); urban microclimate CFM (relative to LES)	Do not pursue	—	Low–Medium (not applicable to transient downburst inflow)
4 Reforecasts / predictability	Lead-time statistics, ensembles around historical events	AWS AIWP (Radford 2025 BAMS 10.1175/BAMS-D-24-0057.1; Pangu/FCNv2 from 2020-10, GraphCast operational from 2022-01, GFS/IFS initial conditions at 00/12 UTC); HENS GMD 18:5575; Vonich arXiv 2504.20238 (initial condition optimization given known future, not forecasting skill)	Candidate: fetch post-2022 multi-event data from AIWP (must first verify archive contains pressure-level variables)	Model-run-free, still requires data retrieval & diagnostics	Medium

What Was Tried and Results

Date	What Was Done	One-Line Result
10-13	Reforecasts with Pangu and FuXi (operational GraphCast training period overlaps the events and the ERA5 version lacks an off-the-shelf wrapper, so FuXi was used instead) for Wister 18 UTC and Akron 06 UTC at lead times of 12/24/48/72/120 h against 13-level ERA5, using diurnally aligned persistence as baseline and three pre-specified criteria	Maximum lead time satisfying all three criteria simultaneously: Wister Pangu 12 h, FuXi 24 h; Akron Pangu 12 h, FuXi none. Parameter by parameter: 0–6 km wind shear error at most lead times is only 0.1–0.6× persistence; CAPE at ≥24–48 h is **worse than persistence** (error ratio 1.5–4). Following the literal wording of the task brief, Criterion 2 averaged across mixed units of J/kg, m/s, and K, so it was effectively dominated by CAPE (a flaw in the task brief's wording, judged strictly as written without post-hoc modification). MLCAPE computed from 13 levels is ~500 J/kg lower than from 37 levels, making vertical resolution itself a major error source. FuXi mirror weights were not verified file-by-file against the original weights
10-13	Added third event ALTU 2023-07-03 18 UTC (supplied by the CM1 track) under the same criteria; also produced a three-event station grid point + 3×3 parameter table	Pangu failed across all five lead times (rejected on MLCAPE alone at every lead time), while FuXi passed only at 12 h; 13-level MLCAPE is 532 vs. 37-level 797 J/kg; model "surface dewpoint" is a proxy taken from the lowest pressure level

Open Items and Next Steps

- 10-13 Intra-track trial rule (10-13 to 10-20): Upon restart, batch pure-QA routine tasks with fixed question banks (abstract initial screening, single-value extraction from papers, table field extraction, initial tag classification) default to ` with 10% audited by Claude to log agreement rate; task types below <90% agreement revert to the original workflow; does not apply to reforecasts, diagnostics, scientific judgment, or review submissions.
- Literature review (default tier) returns → verify citations → ` → finalize assessment table after verdict.
- ideas has returned (10-13 12:30), and Lead has re-verified the parameter-by-parameter error ratios in `metrics.json`. Finalize Row 1 once the ALTU supplemental run returns.
- CM1 has replied (10-13): not currently working on this; offers limited help for their multi-event validation B and serves only as environmental context; provided ALTU 2023-07-03 and the station parameter list → added into the experiment (queued after , criteria unchanged).
- Event definition sources: WRF Lead reply on 10-13; Akron based on card 2DPBW6VX PDF pp. 1–2, 9–11.

- Experimental design note: Parent project review previously canceled the "single-event same-time AI field vs. ERA5" comparison (since AI models are trained on ERA5, that yields no new information). This experiment instead evaluates the growth of environmental parameter errors across **different lead times**, using persistence (ERA5 at initialization time) as the baseline; additionally, event dates must be checked to ensure they fall outside each model's training period (GraphCast/Pangu training ends in 2017, placing Wister 2018 in the test period, pending verification against the original papers).

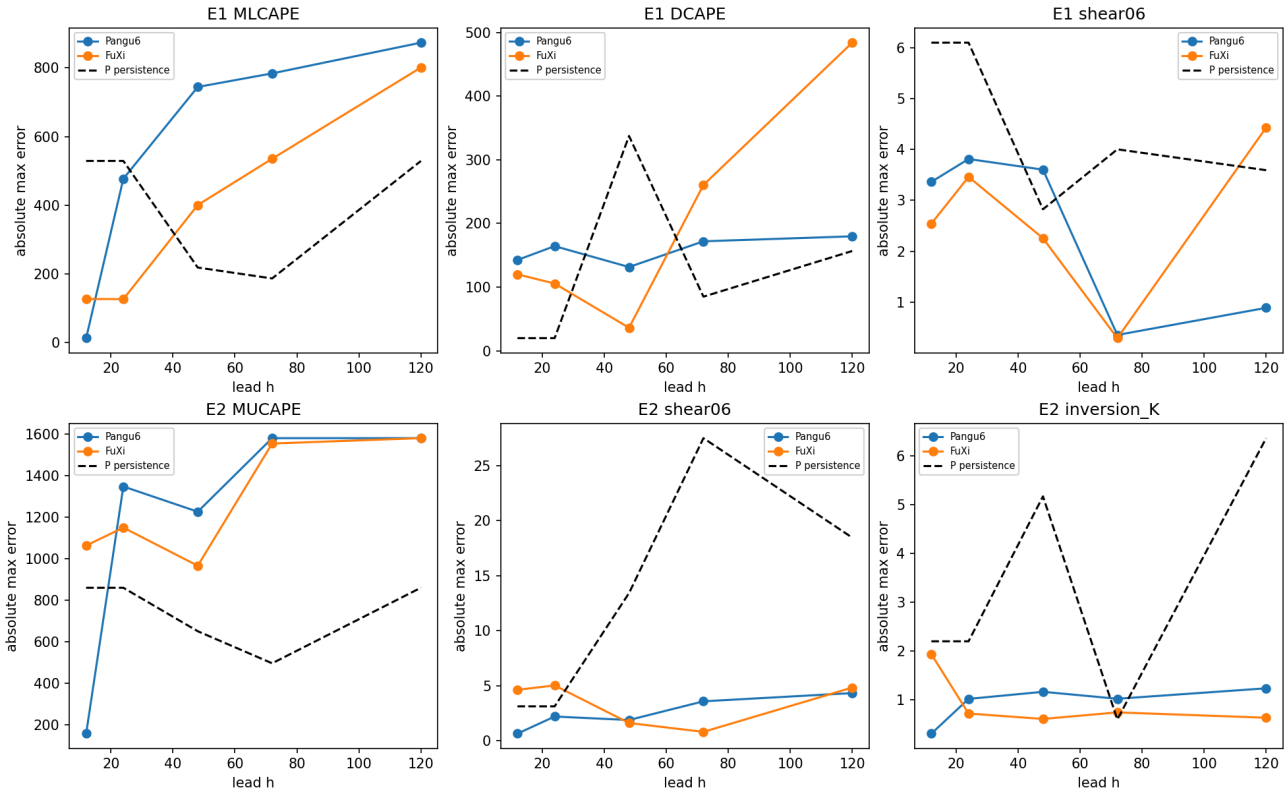
Literature Basis

- 10-13 Showcase page citations have been added to the "Forecasting Uses of AI Weather Models" section (ERA5/ARCO, Pangu, FuXi, GraphCast, earth2studio, ai-models, MetPy, plus the 18 papers underlying the assessment table); reforecast directory added, and new figures with source footnotes have been generated.
- Parent project page "Ancillary Outputs" Row 2b: Feldmann 2024 (arXiv 2406.09474), Hua 2025 GRL (10.1029/2024GL109611), not yet ingested into litwiki cards.
- : 220 m global LES is a **physics-based model**, not AI; abstract-only verification.
- : Generative 3D instantaneous urban wind and temperature fields, abstract only.
- : Physics-constrained network for typhoon near-surface wind profiles, abstract only.
- : AI forecasting + proprietary observation network, superiority claims unverified.
- : Weather model repurposed as a forced climate emulator (arXiv 2605.29976).
- Open source: Row 2b (earth2studio, ai-models under Apache-2.0; licensing terms for Pangu and Aurora weights pending verification).

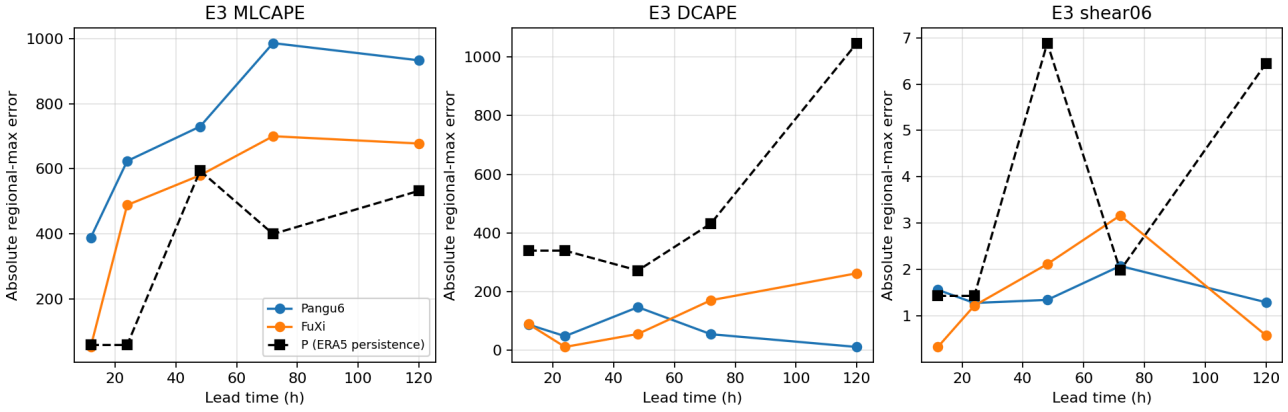
Liang Hu Notes

- 10-13 (key points from Liang Hu's original remarks): Useful for forecasting (especially nowcasting/short-range), capable of providing background fields for tornadoes, thunderstorms, and typhoons; should be even more useful for wind power; asked whether such foundation models also hold reference value for CFD. Approved opening: "Go ahead and open it, you drive it forward."

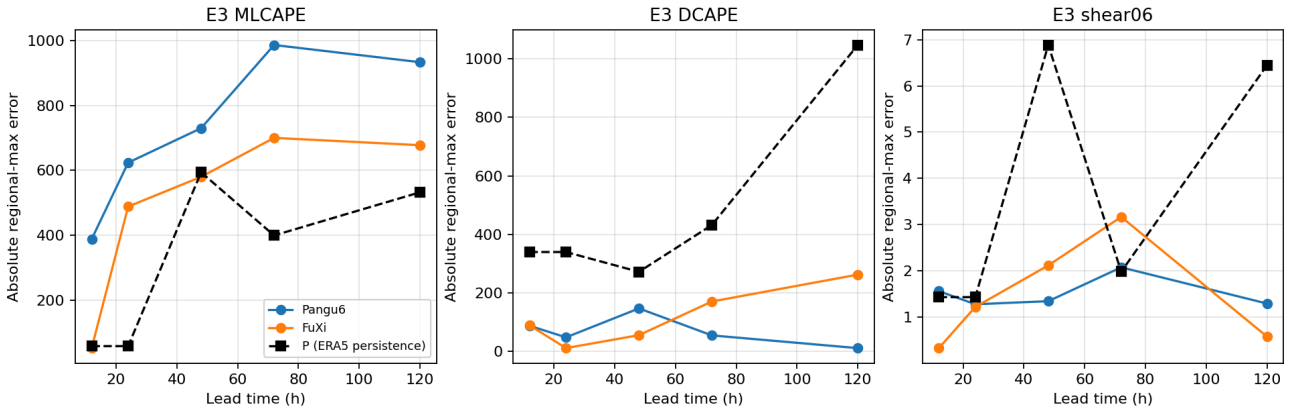
Key figures



errors

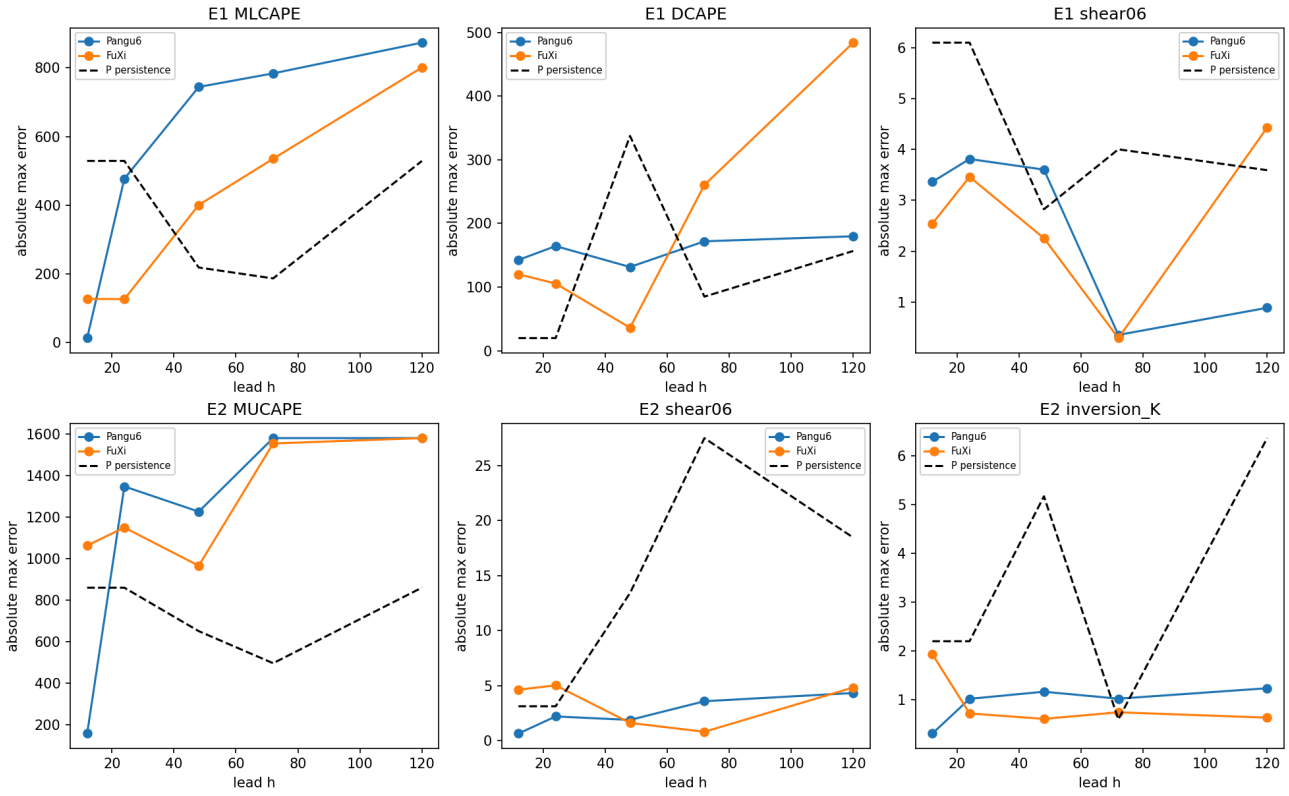


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数据：ERA5（Hersbach et al. 2020, C3S/ARCO）；模型：Pangu-Weather、FuXi（earth2studio）

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数据：ERA5（Hersbach et al. 2020, C3S/ARCO）；模型：Pangu-Weather、FuXi（earth2studio）

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